RESEARCH ARTICLE

Cozie Singapore: Scalable crowdsourced smartwatch micro-surveys to capture longitudinal in-situ urban heat and noise perception

Clayton Miller[a,*], Mario Frei[b] and Yun Xuan Chua[c]

[a]College of Integrative Studies, Singapore Management University, 179873, Singapore;
[b]Berkeley Education Alliance for Research in Singapore (BEARS), 138602, Singapore;
[c]National University of Singapore, 117566, Singapore

ARTICLE HISTORY
Compiled August 2, 2026

ABSTRACT
[PLACEHOLDER: 150–250 word structured abstract. Context (urban sustainability and the human experience of everyday space types); the ORENTH / Cozie-Singapore watch-survey dataset (9,962 responses, 106 participants); the aim (characterize how perception, physiology and ambient weather differ across space types — Home, Office, Classroom, Other Indoor, Outdoor, Transportation); the approach (exploratory analysis + effect-size-based statistical characterization + weather augmentation + discriminative modeling); the headline finding (function/behaviour separates spaces, but ambient weather barely marks space while genuinely tracking thermal preference); and the implication for sustainable urban design.]

KEYWORDS
[PLACEHOLDER: 4–6 keywords — e.g. urban sustainability; thermal comfort; wearable sensing; space-type characterization; watch-survey; Singapore]

1. Introduction

A key component of urban studies is the characterization of how the built environment impacts humans. Often, this analysis uses conventional geospatial information from the data-rich mapping context. These geospatial contexts can then be enriched with a significant amount of metadata through convergence with temporal sensor data, imaging data from satellites or street-view imagery, or from smart devices. Street-view imagery has been used to quantify visual and perceived qualities of the streetscape at scale (Ito, Kang, Zhang, Zhang, & Biljecki, 2024), while network-based methods characterize the morphology and connectivity of the urban fabric (Yap, Stouffs, & Biljecki, 2023). Even human presence itself can be inferred indirectly, for example by estimating building occupancy from mobile phone traces (Barbour et al., 2019).

Despite this effort, significant elements of the human experience in the urban context are missing as observation is not the equivalent of capturing perception. Physical measurements alone are weak predictors of how occupants actually judge their surroundings, because non-physical and psychological factors shape perception (Castaldo et al., 2018), and expectations condition the same measured conditions differently (Schweiker, Rissetto, & Wagner, 2020). Image-based surveys offer one way to elicit

*Corresponding author email: cmiller@smu.edu.sg

perception at scale, asking people to rate qualities such as safety, liveliness, and greenery from street-view imagery (Gu et al., 2025; Ito et al., 2024). However, this perception is elicited from viewing scenes remotely rather than from lived, in-situ experience, so it misses characterizing physical responses from senses beyond sight such environmental and social factors (Quintana et al., 2025). Capturing perception directly has a long methodological basis in ecological momentary assessment and experience sampling, which record subjective states in real time and in context (Christensen, Barrett, Bliss-Moreau, & Lebo, 2003; Stone & Shiffman, 1994), and pairing these reports with location data extends them to the geographically explicit case (Kirchner & Shiffman, 2016).

1.1. Crowdsourcing humans-as-a-sensor data across the urban context

Geographically-explicit ecological momentary assessment (GEMA) developed as a set of methods to record momentary subjective states together with the location in which they occur, and it has been applied across public health and urban analysis (Kirchner & Shiffman, 2016). It has been used to link the daily acoustic environment to real-time noise annoyance (Zhang, Zhou, Kwan, Su, & Lu, 2020) and to relate exposure to natural diversity to momentary mental wellbeing (Hammoud et al., 2024). Standardized architectures and reporting guidelines have since emerged to make these studies more comparable and reproducible (Gharani et al., 2021; Kingsbury et al., 2024). The open-source Cozie platform takes this approach a step further by delivering micro-surveys directly on a smartwatch and pairing each response with the device's onboard physiological and environmental sensors (Tartarini, Frei, Schiavon, Chua, & Miller, 2023). Wearables are increasingly used in this way to capture indoor and environmental quality alongside the people experiencing it (Abboushi et al., 2022), and consumer smartwatches can measure exposures such as ambient noise with useful accuracy (Fischer, Schraivogel, Caversaccio, & Wimmer, 2022).

This study was developed as a follow-on effort to scale up this data collection, with a focus on heat and noise in the urban context. Both exposures are central to the everyday experience of a dense tropical city, where thermal comfort is a persistent concern and noise is a leading source of distraction and dissatisfaction (Appel-Meulenbroek, Steps, Wenmaekers, & Arentze, 2020). Individuals also differ substantially in how they perceive the same conditions, which motivates collecting perception at the level of the person rather than the building (Wang et al., 2018). Linking these reports to geolocated urban context connects them to the digital-twin models increasingly used to represent cities (Ignatius et al., 2024; Miller et al., 2021). The resulting dataset previously supported a public machine-learning competition on predicting heat and noise perception from contextual data (Miller, Ibrahim, et al., 2025; Miller et al., 2023), and this paper instead characterizes the collected experience itself across the range of everyday urban spaces.

2. Methods

This study characterizes human-experience data collected in Singapore with the Cozie platform, an open-source application for the Apple Watch that administers watch-based micro-surveys and records the wearer's onboard physiological and environmental sensors (Tartarini et al., 2023). The overall approach was first proposed as a smartwatch-driven just-in-time intervention framework for building occupants (Miller, Chua, Frei, & Quintana, 2022), and the deployment and its instrumentation have since been described in methods papers that also report preliminary results (Miller, Ibrahim, et al., 2025; Miller et al., 2023), with the same platform underpinning related work on just-in-time interventions for heat and noise (Miller, Chua, et al., 2025). The design is in established practice for ecological momentary assessment and its geographically-explicit extension, including recommendations for sampling, scheduling, and transparent reporting (Kingsbury et al., 2024; Reichert et al., 2020). Because the platform continuously records location and physiological signals, the deployment also considers the issues raised for mobile-sensing studies (Fuller, Shareck, & Stanley, 2017). The remainder of this section describes the field study and participants, the structure of the micro-survey, the linked wearable, weather, and urban-context data, and the processing steps that produce the survey-level dataset used for the characterization that follows.

2.1. *Crowdsourced watch-based micro-survey and physiological and environmental data*

The field study reported here is the same deployment described in detail by Miller, Chua, et al. (2025), and the following is a summary of that protocol, illustrated in Figure 1. Participants were recruited through the National University of Singapore (NUS) Student Work Scheme and by word of mouth, producing a pool made up mostly of NUS students together with a smaller number of adults from outside the university. The cohort was predominantly young, with roughly 73% aged 18 to 30, and comprised 39% male and 61% female participants, and ethical approval was granted by the NUS Institutional Review Board. Each participant used an Apple Watch running the Cozie application paired with an iPhone, and loan devices were provided to those who lacked a compatible watch or phone. Participants were asked to wear the watch during working hours, at least from 9 AM to 7 PM on weekdays, for approximately four weeks. They were instructed to complete at least 100 micro-surveys together with weekly surveys and an in-person exit interview. Some participants provided responses outside the parameters in these guidelines and those submitted within Singapore were retained.

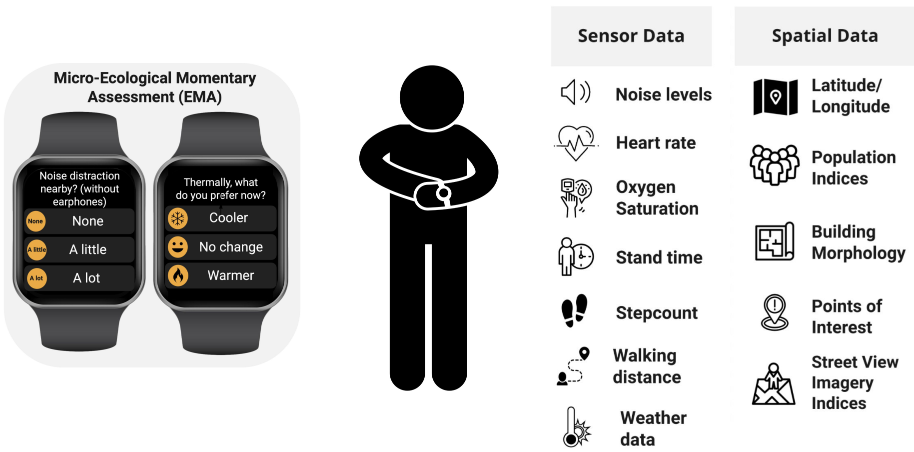

Figure 1. Cozie Singapore study design linking crowdsourced smartwatch micro-surveys (left) to the wearable-sensor and urban spatial data each response is joined to (right).

The deployment ran in two phases that differed in how just-in-time intervention messages were triggered. As summarised in Figure 1, each watch-based micro-survey response is time-stamped and geolocated, and is paired with the wearer's onboard physiological signals and with linked outdoor weather and urban-context data. The present analysis draws on a slightly different prepared dataset from this deployment, comprising 9,962 micro-survey responses from 106 participants, and focuses on characterizing the reported experience rather than the intervention outcomes.

Each micro-survey is a short set of questions answered directly on the watch face, shown in Figure 2. A response records the participant's thermal preference (*Cooler*, *No change*, or *Warmer*) and the level of nearby noise (*None*, *A little*, or *A lot*). When noise is present, follow-up questions capture the kind of noise and whether earphones are being worn. The survey also records the reported location or space type, whether the participant is *Alone* or in a *Group*, and the activity being performed, with the activity options depending on whether the participant is alone or with others. Several of these questions are conditional, so later items such as noise kind and activity are answered only for the responses that reach them.

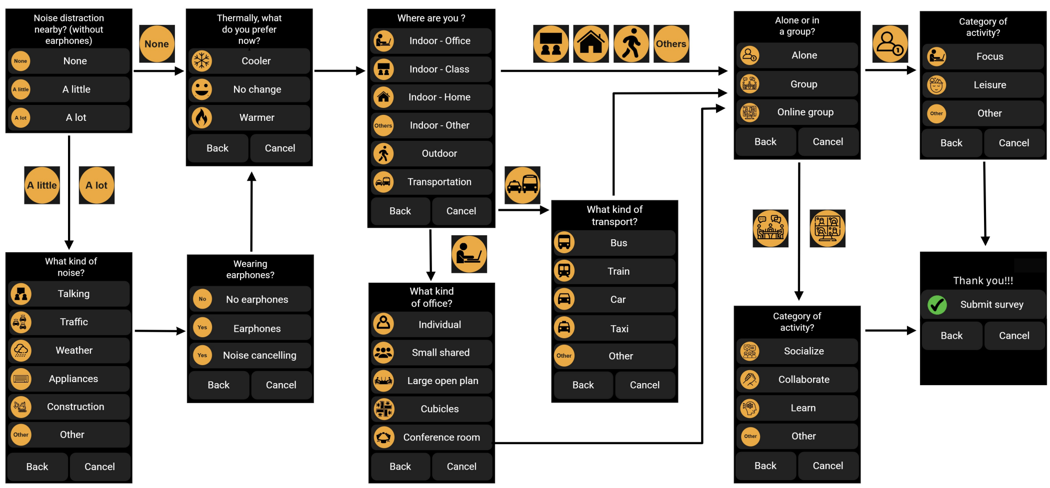

Figure 2. Smartwatch micro-survey interface showing the left-to-right flow of watch screens from the core questions and their conditional follow-ups to submission.

Each geolocated response was linked to a set of urban-context indicators derived with Urbanity, an open-source tool that builds feature-rich urban networks from open geospatial data (Yap et al., 2023). These indicators summarise the built environment around each response location, including streetscape view indices for greenery, sky, buildings, and roads, together with visual complexity, building count, population, and nearby points of interest across categories such as commercial, food, and recreational uses. They provide the objective urban characterization used to test whether physical context adds information beyond the self-reported experience.

2.2. Data processing and characterization

The raw deployment data combines high-frequency watch and health streams with the sparser watch-survey responses in a single time-indexed record. Survey responses were first screened to those triggered on the watch that carried valid, non-zero coordinates falling inside the Singapore bounding box, which produced the 9,962-response analysis set. For each retained response, the seven temporal watch-health streams were aggregated over three pre-survey windows of six hours, one hour, and ten minutes, summarising each window by its count, mean, spread, and extremes, and by an accumulated total for step count, walking distance, and stand time. The nearest weather-station observations were aggregated over the same three windows, and a heat index was derived from air temperature and relative humidity. These health and weather aggregates were then merged back onto the survey rows and checked for consistency, yielding one analysis-ready record per response that carries the original survey and location fields alongside the linked physiological, weather, and urban-context features used throughout the analysis.

The reported experience was then characterized against the reported space type using the analysis-ready record described above. Each response was assigned to one of the space types, and each categorical perception variable was cross-tabulated with the space type to form the contingency tables used below. Conditional questions such as noise kind and earphone use have lower coverage, so their profiles describe only the responses for which the follow-up was observed. Because responses are clustered within participants, the significance markers reported below are treated as descriptive screens, and effect sizes and the coherence of contrasts across related variables are used as the primary read-outs. The three methods below produce the perceptual characterization used for analysis.

2.2.1. *Conditional perception fingerprints*

For each space type s and response category r, the conditional proportion is the maximum-likelihood estimate of the categorical distribution within that space,

$$\hat{P}(\text{response} = r \mid \text{space} = s) = \frac{n_{rs}}{n_s}, \tag{1}$$

where n_{rs} is the number of responses of category r in space s and n_s is the total number of responses in that space. Each row therefore describes a single space's own response profile and sums to one within each survey variable, which is the fingerprint shown in Figure 13. No smoothing or shrinkage is applied, so cells based on few responses are noisier than their shading suggests.

2.2.2. *Relative distinctiveness through lift*

To separate what is distinctive to a space from what is simply common everywhere, each conditional proportion is divided by the corresponding corpus-wide proportion to give the lift,

$$\text{lift}(r, s) = \frac{P(\text{response} = r \mid \text{space} = s)}{P(\text{response} = r)}. \tag{2}$$

A value of one indicates a response that is as common in the space as in the corpus overall, values above one indicate over-representation, and values below one indicate under-representation, as shown in Figure 14.

2.2.3. Association strength and cell-level contrasts

The overall association between space type and each perception was tested with a Pearson chi-square statistic on the $r \times k$ contingency table,

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \tag{3}$$

where O_{ij} and E_{ij} are the observed and expected counts under independence. Because χ^2 grows with sample size, it was rescaled to a bounded effect size using Cramér's V,

$$V = \sqrt{\frac{\chi^2}{n\,(\min(r,k) - 1)}}, \tag{4}$$

interpreted with conventional descriptive bands (negligible < 0.1, weak 0.1–0.2, moderate 0.2–0.3, strong > 0.3). To identify which individual space and response combinations drive the association, adjusted standardized residuals were computed for each cell,

$$z_{ij} = \frac{O_{ij} - E_{ij}}{\sqrt{E_{ij}\,(1 - r_i/n)\,(1 - c_j/n)}}, \tag{5}$$

where r_i and c_j are the row and column totals and n is the grand total. Under independence each z_{ij} is asymptotically standard normal, so cells with $|z| > 1.96$, 2.58, and 3.29 exceed the conventional 0.05, 0.01, and 0.001 thresholds and mark the space and response pairs shown as over- or under-represented in Figure 15.

3. Results

The collected data support the analysis of the volume, coverage, and timing of the responses, to the distributions of the survey and its linked physiological and weather measurements, and then to how the reported experience differs across the everyday spaces in which it was recorded.

3.1. Data collection, coverage, and temporal characteristics

Figure 3 shows the mapping of the responses across Singapore. The densest grid cells sit over residential estates and the NUS university campus, consistent with the roughly 45% of responses reported from home, while coverage thins quickly toward the periphery even as individual responses still reach the central business district and points along the transport network. The coverage is therefore broad but uneven, anchored to where participants live and work.

Figure 4 illustrates the breakdown of data collection across the cohort of 106 participants. Contribution is spread evenly enough that no small group dominates the dataset, with a median of 98 responses per person, close to the 100-survey target, and the most active tenth of participants supplying only about 12% of the total. The range from 17 to 158 responses does mean that some participants are somewhat unequally

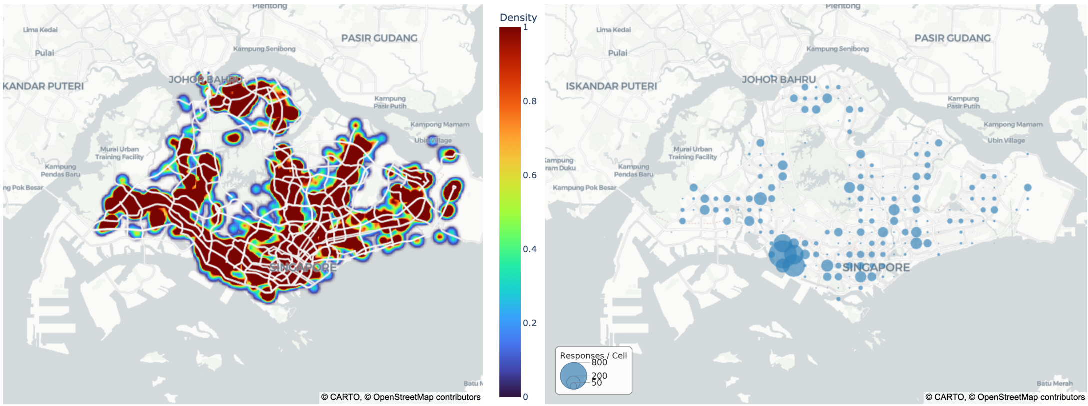

Figure 3. Spatial coverage of survey responses across Singapore, shown as a kernel-density surface of response locations ($N = 9{,}931$; left) and response counts aggregated into approximately 1-km grid cells (276 cells, top cell 897 responses; right).

represented, the analyses that follow rely on within-space proportions and effect sizes rather than raw response counts.

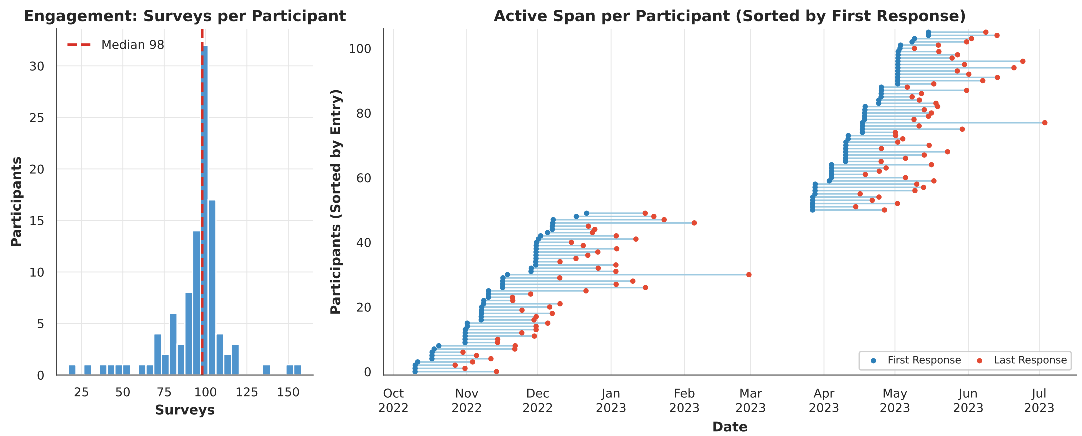

Figure 4. Participant engagement over the deployment: the distribution of surveys per participant (left; median 98) and each participant's active span from first to last response (right).

The temporal cadence in Figure 5 reflects a protocol built around the working day. Responses concentrate sharply between 9 AM and 7 PM and peak in the mid-afternoon, and weekdays account for all but about 17% of the data, so the dataset captures waking, active hours far more than nights or weekends. The calendar-week panel shows the response volume rising and falling with the phased recruitment of participants.

Timing is normalised within each space type in Figure 6, so it reads as a set of temporal signatures for when each kind of space is used. Offices and classrooms are almost only weekday, daytime settings, with about 96% of office responses on weekdays, while home stretches into the evening and holds the largest weekend share of any space. The commute spaces give the sharpest signatures, as outdoor and transportation responses break from the mid-afternoon indoor peak and rise in the early evening around 18:00 to 19:00. Timing alone already begins to separate the spaces before any perception is considered.

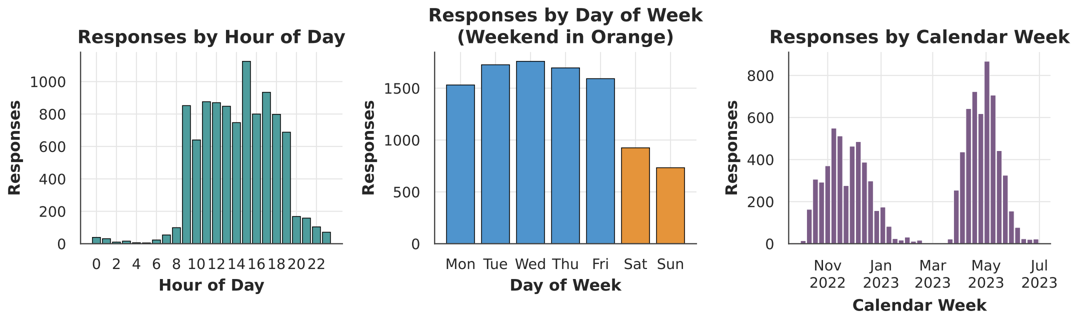

Figure 5. Temporal cadence of survey responses in Singapore local time by hour of day (left), day of week (middle), and calendar week (right).

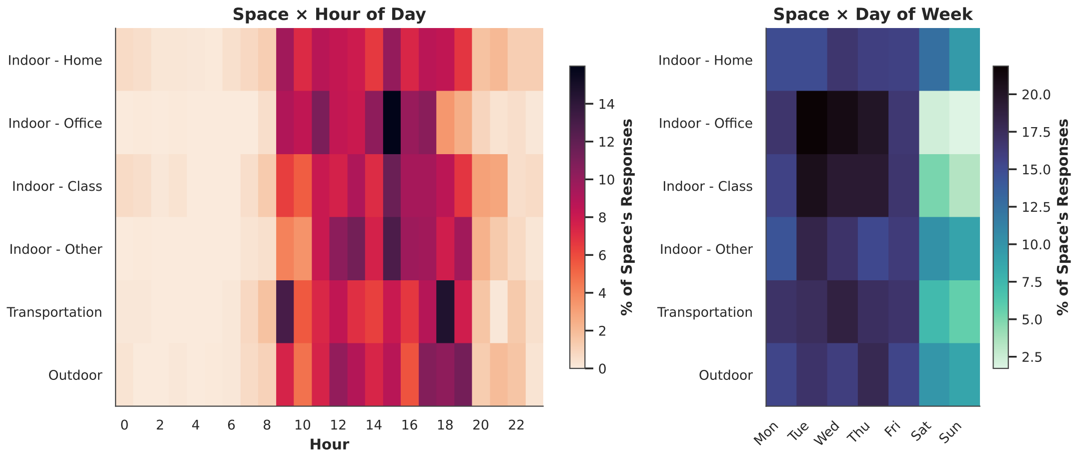

Figure 6. Row-normalised response timing by space type, showing each space's share of responses across hour of day (left) and day of week (right).

3.2. Survey responses and linked temporal data coverage

Figure 7 gives the overall distribution of every question, and read together the panels sketch the typical moment a participant reported. Thermal preference is strongly asymmetric. A small majority want *No change* (54%), almost all of the rest want to be *Cooler* (40%), and *Warmer* is rare (6%). This is the expected pattern for a warm, humid climate, though it also reflects that many indoor spaces are air-conditioned rather than exposed to the outdoor weather. Perceived noise is more balanced, with roughly equal shares reporting *None* (46%) and *A little* (44%) and only about one in ten reporting *A lot*. Noticing some noise is the norm, while strong distraction is uncommon. Where noise is present, the follow-up question attributes it most often to people talking (44% of noise-kind responses), well ahead of appliances and traffic, so the main acoustic nuisance in this cohort is human rather than mechanical.

The other panels place these responses in context. Most responses come from home (45%), participants are usually *Alone* (72%) rather than in a *Group*, and the activity questions show focused work and leisure in solitary moments and socialising in group ones. Earphone use is uncommon even though the question is reached in about half of surveys, as most of those responses report *No earphones* and active noise cancelling is rare. Participants therefore seem to tolerate rather than mask their sound environment.

The coverage panel shows why the linked analysis is feasible, since the one-hour pre-survey window carries weather for about 96% of surveys and heart rate, movement, and measured noise for roughly 90%. Nearly every self-report has matching sensor data.

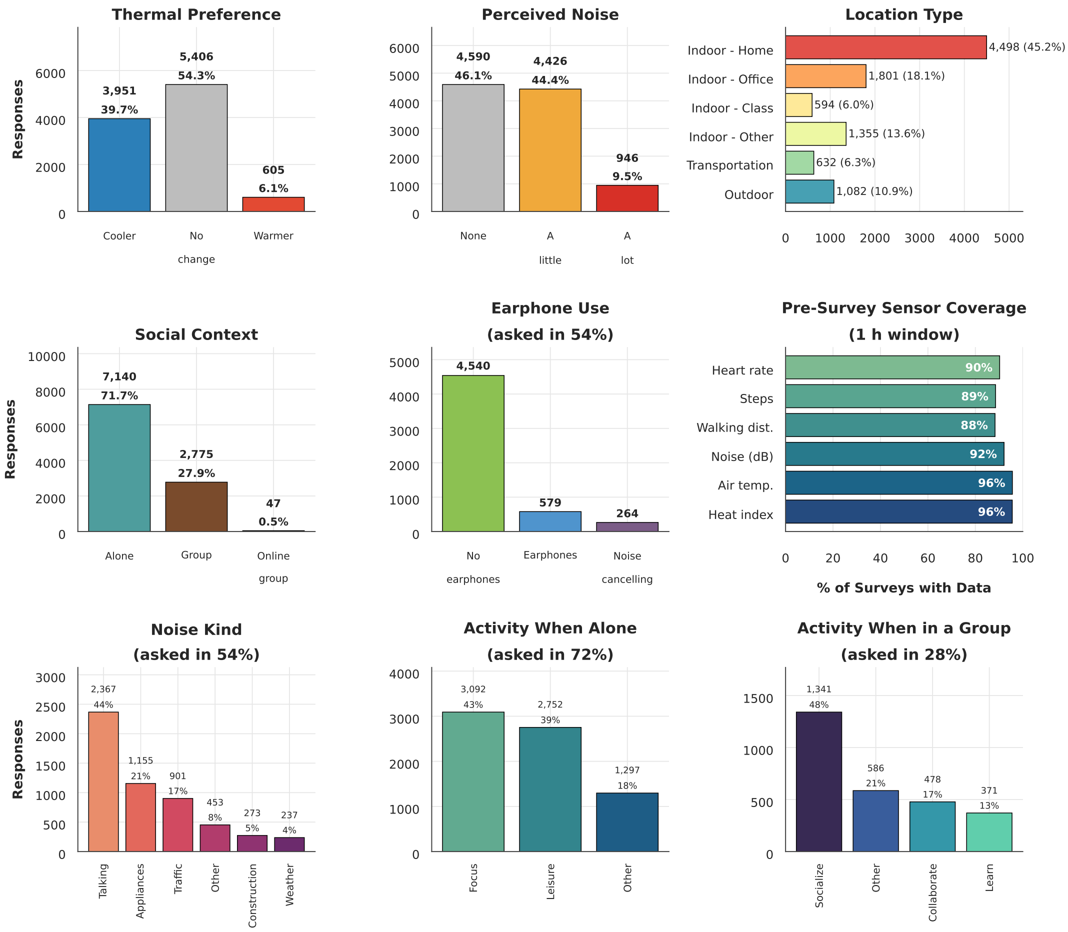

Figure 7. Response distributions for each survey item and pre-survey linked-sensor coverage (N = 9,962 responses, 106 participants).

The same responses appear as a flow through the questions in Figure 8, which makes the conditional structure clearer than the separate bars can. Reading from left to right, each answer splits into the next question, the conditional branches such as noise kind and earphones carry only the half of responses that reach them, and activity divides by whether the participant is *Alone* or in a *Group*. The ribbon widths are drawn to scale, so the proportion of every category can be compared in a single view. The common home, solitary, and *No change* or *Cooler* responses form the thick central spine, and the rarer categories taper into thin branches.

The linked wearable-health streams appear in Figure 9, each summarised over the six-hour, one-hour, and ten-minute windows before a survey. The window matters most for the movement metrics. Median step count and walking distance fall as the window narrows, from about 85 to 42 steps per reading between the six-hour and ten-minute windows, because a short look-back captures only the last few minutes of activity. Heart rate is far more stable at roughly 80 bpm across all three windows. Coverage

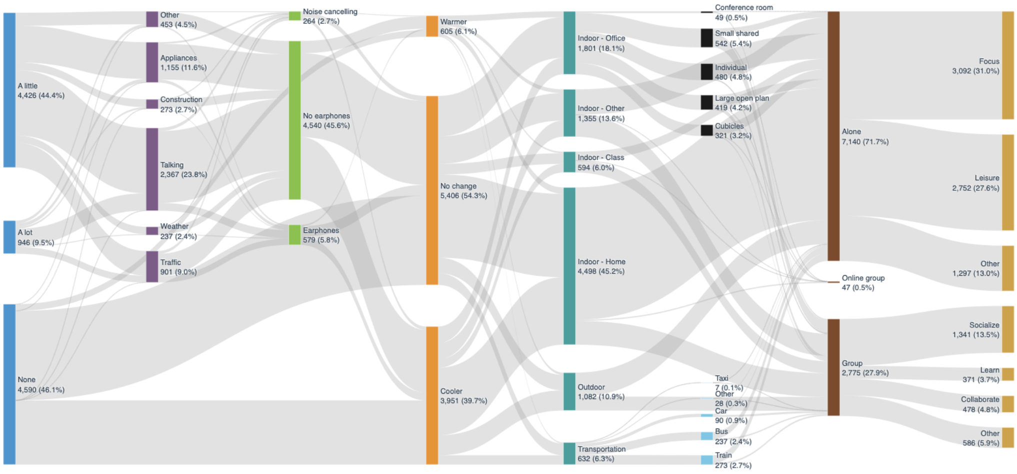

Figure 8. Sankey diagram of the conditional response flow through the questionnaire, with band width proportional to the number of responses along each path.

moves the other way and limits how tight the window can be. The six-hour window carries movement and noise for over 95% of surveys, but at ten minutes audio coverage falls to about a third and step coverage to little more than half. Resting heart rate and blood-oxygen streams are sparse in every window. The one-hour window is used as the working aggregate throughout, since it keeps coverage near 90% while staying close to the moment of the survey.

Figure 10 summarizes the nearest-station outdoor weather over the same windows and its distributions are narrow and stable, as expected for an equatorial climate. Air temperature has a median near 29 °C and relative humidity near 73%, which give a heat index around 34 °C that changes little across the windows. The outdoor thermal environment is consistently warm whatever the look-back length. Rainfall is the one strongly window-dependent feature because it accumulates over time, with about 28% of six-hour windows containing some rain against only 5% of ten-minute windows. These values describe outdoor conditions at a station a median of roughly 3 km from the response. As noted above, they are not the participant's actual microclimate, which is often indoor and air-conditioned.

3.3. Thermal preference and noise characterization

The two dominant strands of the reported experience, wanting to be cooler and being distracted by noise, were examined before they are related to space type. Cooling preference against outdoor heat and across participants is characterized in Figure 11, and the demand is real but only loosely tied to the outdoor thermometer. Across all responses, the share of *Cooler* answers rises with outdoor heat, from about a third when the pre-survey heat index is near 30 °C to roughly half above 37 °C. This is the expected direction for a tropical setting, though the outdoor reading is only a proxy for the participant's actual microclimate, which is often a controlled indoor space rather than the outdoor weather. The link is much weaker between people than within the heat range. A participant's mean outdoor temperature explains little of who wants cooling ($r \approx 0.33$), and the per-participant panel is very wide, with the *Cooler* share

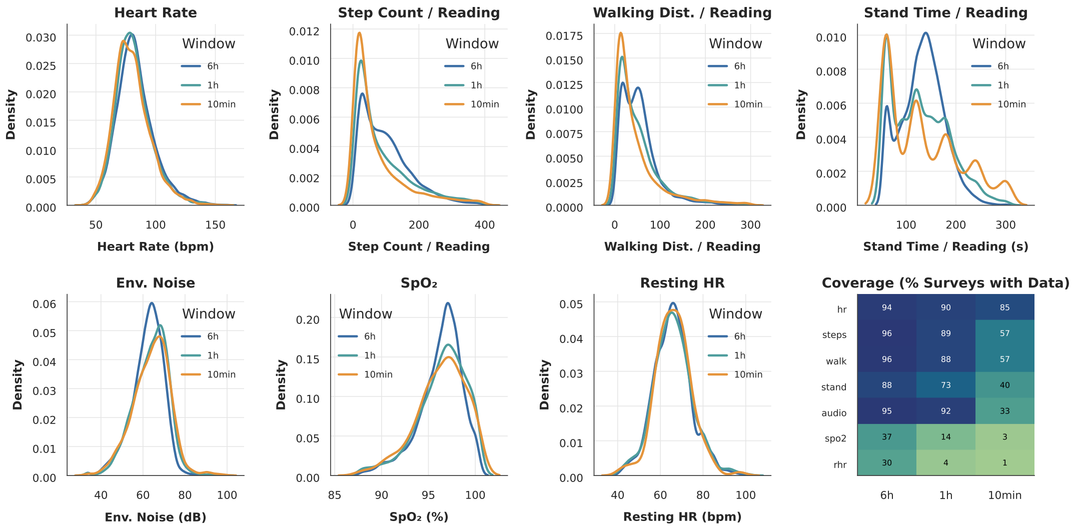

Figure 9. Distributions of pre-survey wearable-health metrics across the 10-minute, 1-hour, and 6-hour aggregation windows: seven wearable health and audio streams as overlaid per-window density curves, plus a metric-by-window data-coverage heatmap.

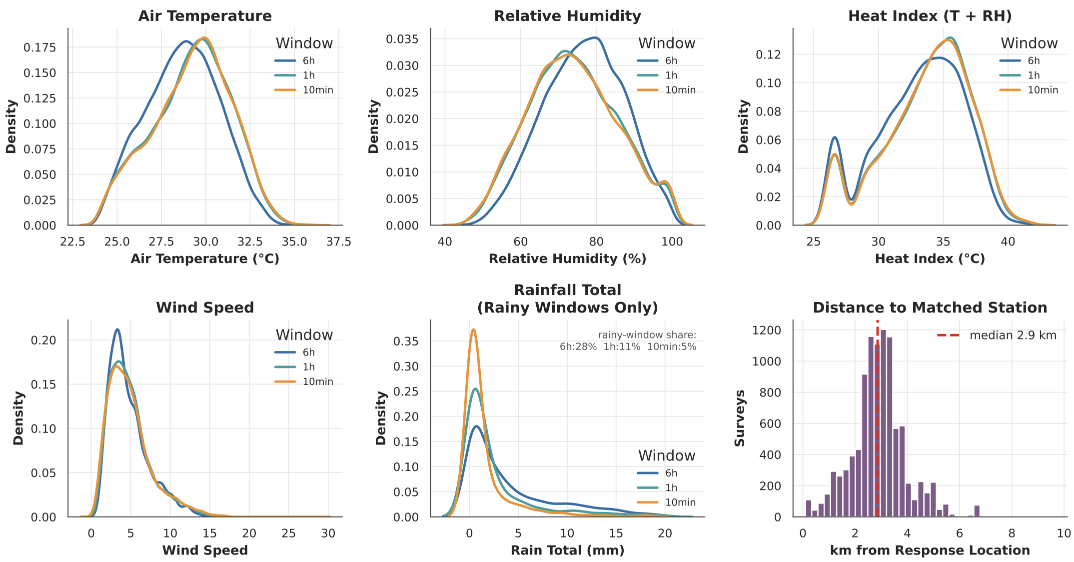

Figure 10. Nearest-station outdoor weather across the 10-minute, 1-hour, and 6-hour pre-survey windows, with distributions for five ambient metrics and the distance to the matched station (median 2.9 km).

running from under 10% for the most heat-tolerant participants to nearly 80% for the most cooling-seeking. This spread is the main point of the figure. Cooling preference is as much a personal trait as a response to measured conditions, and this is the individual variation that person-level sensing is designed to capture.

The same characterization for the sound environment is given in Figure 12, using the linked one-hour sound-level measurement, and here the self-reports and the sensor agree well. Measured sound level rises steadily with perceived noise, from a median near 62 dB for *None* to 66 dB for *A little* and 70 dB for *A lot*. The separation is consistent despite the coarse three-point scale. The measured levels also track where people are, with home and office the quietest settings at around 62 dB while other-indoor, transportation, and outdoor environments sit near 69 to 70 dB, and the sources most often reported as loud, traffic and talking, carrying the highest levels. Together the two figures frame the analysis that follows. Thermal preference is a personal signal that only loosely tracks the surrounding environment, while noise perception is tied to a measurable exposure that varies with the space around the participant.

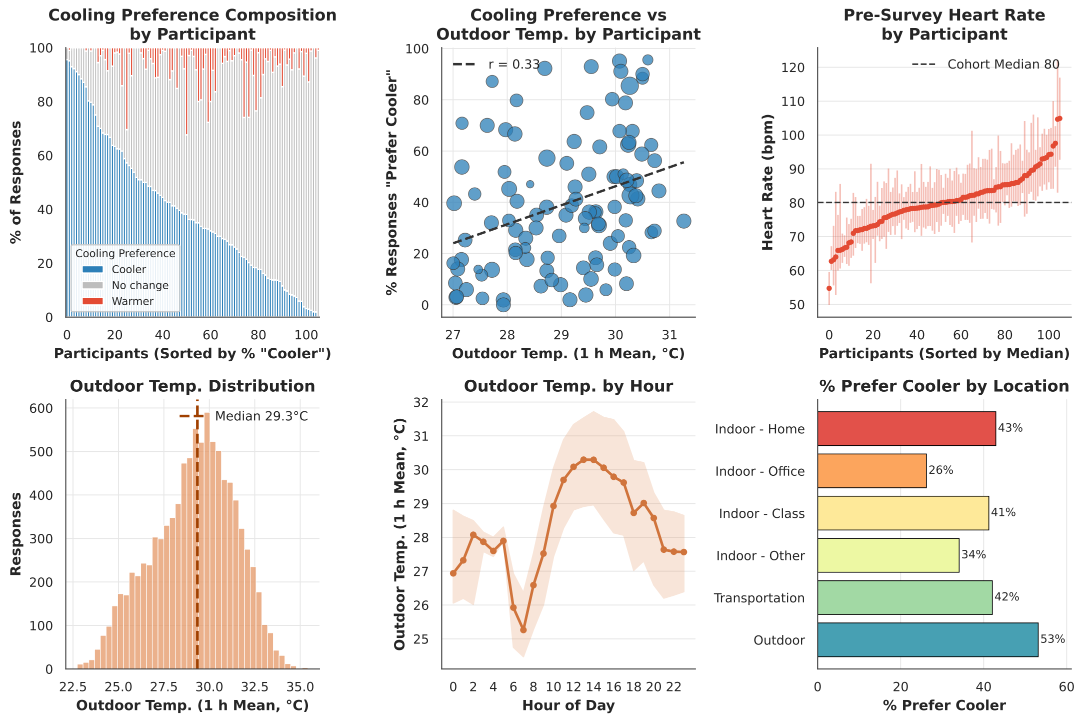

Figure 11. Participant-level and contextual patterns in cooling preference, relating per-participant composition, outdoor temperature, heart rate, and location type.

3.4. Perceptual fingerprints and association with space type

With each perception characterized on its own, the analysis now asks how the reported experience differs across the spaces in which it was recorded. It does so in three layers that build on one another. The first layer, in Figure 13, reads each space type as a profile by normalising within its own row. Every cell gives the probability of a response given the space, so each row describes the mix of experiences in that setting rather than how many responses it contributed. Read this way, the spaces separate

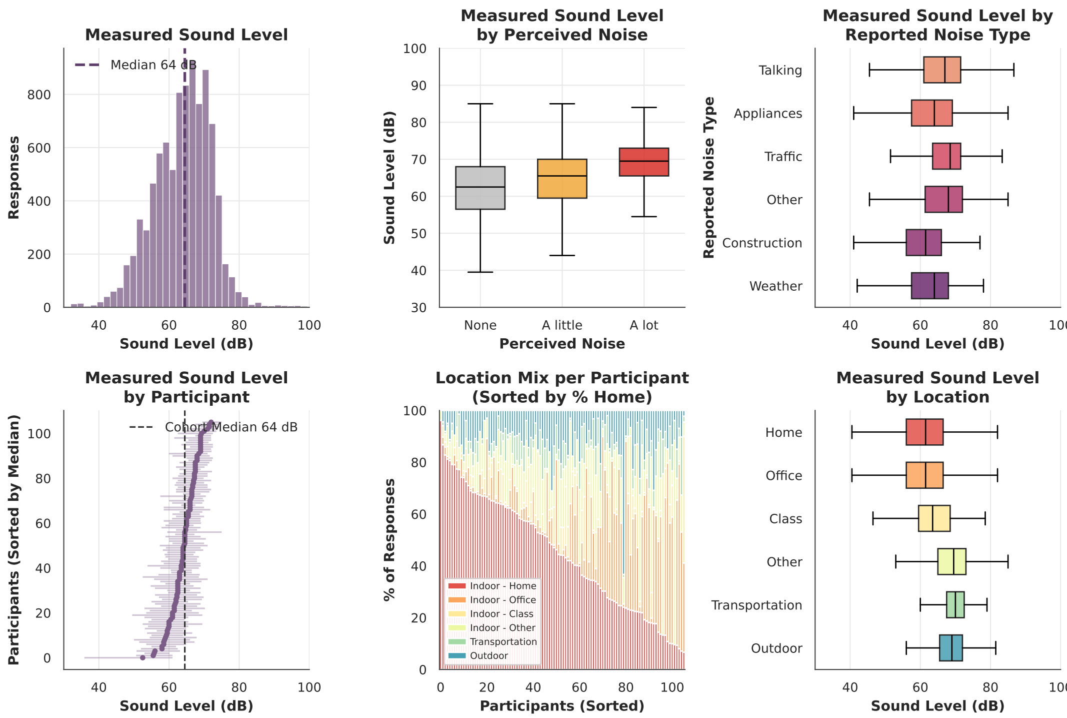

Figure 12. Measured sound level (1-hour pre-survey mean) summarised across subjective, participant, and location contexts.

into recognisable characters. Home is a quiet, solitary setting, with most responses reporting *None* for noise and about 84% reporting *Alone*, while classrooms and other-indoor spaces are mostly social. The commute spaces stand out for discomfort, with the highest shares of *A lot* of noise, reaching roughly one in four responses in transportation against only about 3% at home. Thermal preference shifts more gently, from only about a quarter of office responses wanting *Cooler* to just over half of outdoor responses. This fits the heavily air-conditioned office and the exposed outdoor setting, and it is a reminder that thermal preference reflects the immediate controlled environment more than the outdoor weather. The fingerprint is intuitive but has one limit. Because some responses are common everywhere, a high within-space probability does not by itself mean the response is characteristic of that space.

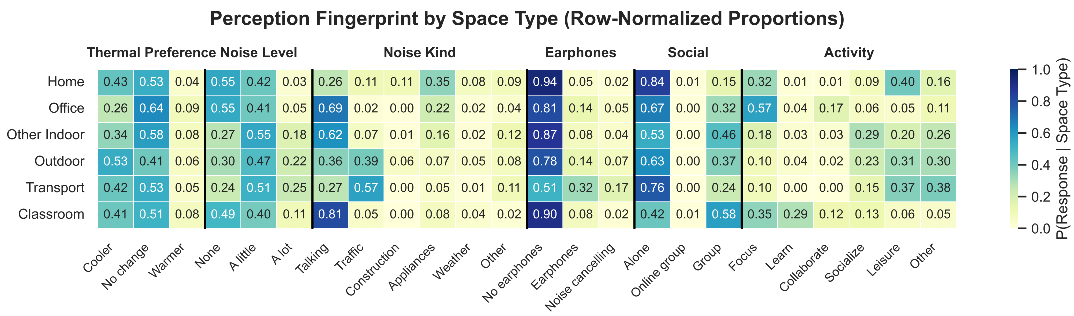

Figure 13. Conditional perception fingerprint by space type in a heatmap of $P(\text{response} \mid \text{space type})$ with space types in rows and response options (grouped by survey dimension) in columns.

The second layer, in Figure 14, divides each within-space probability by the response's overall proportion in the dataset. A lift above one marks a response that is more common in a space than in the data as a whole, and a lift below one marks one that is rarer. This turns a profile into a signature, because it isolates what is distinctive to a space rather than what is simply frequent. The contrast sharpens the same story and adds detail the fingerprint hides. *A lot* of noise is over-represented by about 2.6 times in transportation and 2.4 times outdoors, *Group* is over-represented most in classrooms (lift near 2.1), and *Warmer*, a rare response overall, is concentrated in the office (lift near 1.6) where air-conditioning can overshoot. Lift also makes the source of noise a spatial fingerprint in its own right, as traffic accounts for most reported noise in transportation and a large share outdoors, while talking dominates the noise reported in classrooms and offices. The limit of lift mirrors that of the fingerprint. It flags distinctive combinations but does not show whether any single contrast is larger than chance would produce.

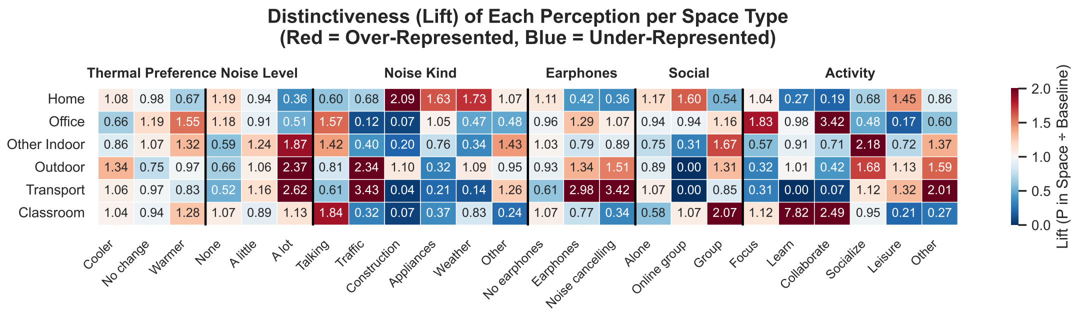

Figure 14. Relative distinctiveness of perceptions by space type, shown as lift, $P(\text{response} \mid \text{space type})/P(\text{response})$, with space types in rows and response options (grouped by survey dimension) in columns.

Figure 15 illustrates the third layer of analysis that adds the statistical dimension the first two lack. It compares each space and response combination against what independence would predict, using adjusted standardized residuals whose sign gives the direction of the effect and whose size is set on a standard-normal scale. This layer confirms which distinctive contrasts are large relative to sampling noise, and it agrees with the overall effect sizes. The association with space type is strongest for what a person is doing and hearing, with Cramér's V near 0.38 for group activity, 0.35 for solitary activity, and 0.28 for the kind of noise, and weakest for thermal preference at about 0.12. The residuals make the behavioural picture explicit, as transportation and outdoor spaces stand out for loud, traffic-driven noise, classrooms and offices for talking and for group or focused activity, and home for quiet, solitary responses. Thermal preference produces only muted residuals scattered across spaces. Because the responses are clustered within 106 participants, the significance markers are read as descriptive screens rather than formal tests, and the agreement of all three layers on the same ordering is the stronger evidence. The lesson across the three layers is that space type is written far more clearly into what people are doing and hearing than into how warm they feel. This weak thermal signal is consistent with people often being in controlled indoor settings rather than exposed to the outdoor weather, and it is the central characterization result of the study.

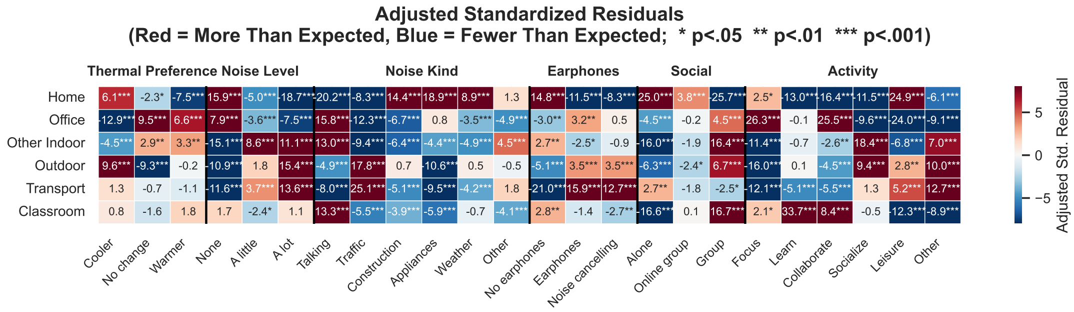

Figure 15. Adjusted standardised residuals for space-type by response combinations, where red cells occur more often and blue cells less often than expected under independence.

4. Discussion

The three analytical layers converge on a coherent characterization of how urban space types differ in reported experience. The clearest separations are behavioural and acoustic rather than thermal: what people are doing and hearing identifies a space far more sharply than how warm they feel. Tables 1 and 2 collect the findings that reinforce prior expectation and establish this baseline character of each setting, grouped by the acoustic environment and by social context and thermal comfort. Each entry pairs a short insight with the supporting detail drawn from the fingerprint, lift, and residual analyses of Section 3.4.

Table 1. Space-type findings that reinforce prior intuition: the acoustic environment.

Key insight	Detail
Commute spaces are the noisiest	Transportation and outdoor responses carry the highest *a lot* noise shares (≈ 25% and 22%), against only about 3% at home.
Home and offices are the quietest settings	Both sit near a 62 dB median sound level, the lowest among the measured spaces.
Perceived loudness tracks the setting	Louder spaces attract more *a lot* noise reports, so subjective loudness follows the physical environment.
Traffic defines commute noise	Traffic is the dominant reported noise source in transportation (≈ 57%) and outdoors (≈ 39%).
Talking defines indoor noise	Talking dominates the acoustic complaints of classrooms (≈ 81%) and offices (≈ 69%).

These reinforcing findings matter less as surprises than as evidence that a watch-based micro-survey recovers the settled structure of urban experience that the field has documented through other instruments. Urban research has increasingly treated residents' perceptions of their surroundings as a primary measurement in their own right rather than as a noisy proxy for physical conditions, on the argument that how people read an environment can matter more for outcomes such as self-rated health than the measured environment itself (Kim, 2016). Capturing those perceptions at the scale of a city has motivated a parallel line of mobile-crowdsourcing work, in which residents contribute geolocated subjective reports—of safety, comfort, or nuisance—from their own devices as they move through the urban fabric (Phuttharak & Loke, 2020). The present study sits within this tradition and extends it by delivering the survey on

Table 2. Space-type findings that reinforce prior intuition: social context and thermal comfort.

Key insight	Detail
Home is overwhelmingly solitary	About 84% of home responses report being *alone*, the most solitary setting measured.
Classrooms are the most social setting	Classrooms are the most group-oriented space, with a group lift near 2.1.
Activity is the strongest spatial marker	Group activity is the response most tightly associated with space type (Cramér's V ≈ 0.38).
Offices show the lowest cooling demand	Only about 26% of office responses want *cooler*, consistent with heavy air-conditioning.
Outdoor spaces show the highest cooling demand	Just over half ($\approx 53\%$) of outdoor responses want *cooler*.

the wrist and pairing each response with onboard sensing, so that the solitary-quiet character of home and the social character of classrooms emerge from in-situ reports rather than from recalled or remotely elicited judgements.

The acoustic results align closely with what this body of work has found through other means. That commute spaces carry the highest *a lot* noise shares and that traffic dominates their reported noise source echoes mobile-exposure studies of commuters, in which personal exposure to traffic-driven noise and air pollution is highly context-dependent and is best captured by combining wearable measurement with in-situ, on-the-move accounts (Marquart, Schlink, & Nagendra, 2022). It is equally consistent with spatio-temporal urban assessments that locate the loudest ambient conditions in traffic-dense cores rather than in the residential or green periphery (Kumar & Pandey, 2013). The convergence of a wrist-based micro-survey with dosimetry-, interview-, and remote-sensing-based approaches on the same ordering of spaces is more informative than any single instrument would be, and it is the kind of cross-method corroboration that motivates treating the quieter findings that follow as trustworthy.

Methodologically, the layered reading of subjective, physiological, and spatial signals used here belongs to a growing strand of multi-method urban perception research, exemplified by integrations of participatory mapping, neurophysiological response, and self-reported emotion to characterise experiences such as perceived walkability (Yıldırım, Heyik, Güleç Ozer, & Sungur, 2026). Within the specific pairing of thermal and acoustic experience, the watch-based platform used here has already supported city-scale prediction of noise distraction and thermal preference (Miller, Ibrahim, et al., 2025), and the characterization below establishes the descriptive structure on which such predictive work rests.

Beyond this expected picture, several results are less obvious and reshape how the characterization should be read. Thermal preference, the signal one might expect a tropical climate to dominate, is in fact the weakest spatial marker, while distinctiveness that raw proportions understate emerges only once responses are weighed against their overall frequency. Tables 3 and 4 gather these counterintuitive and analytical findings, separating the environmental and thermal surprises from the behavioural signatures and from the framing that effect sizes, rather than p-values, provide under participant nesting.

Table 3. Counterintuitive environmental and thermal signals.

Key insight	Detail
Thermal preference is the weakest spatial signal	Its association with space type (Cramér's V ≈ 0.12) sits far below activity or noise.
Cooling demand separates spaces only modestly	Outdoor demand ($\approx 53\%$) exceeds home ($\approx 43\%$) by less than an air-conditioning-driven intuition would suggest.
Offices can over-cool	Wanting *warmer*, rare overall, concentrates in offices (lift ≈ 1.6), a signature of air-conditioning overshoot.
The kind of noise matters more than the amount	The source of noise fingerprints a space more sharply than how loud it is.
Lift exposes hidden distinctiveness	*A lot* of noise is $\approx 2.6\times$ over-represented in transportation relative to the whole corpus, a contrast raw proportions understate.

Table 4. Behavioural signatures and analytical framing.

Key insight	Detail
Behaviour separates spaces best	What a person is doing distinguishes space types more sharply than any environmental variable.
Home is distinctively, not just frequently, solitary	Even against an already alone-dominated corpus, home's alone lift exceeds one.
Commuting is loud and lonely	Transportation pairs loud, traffic-driven noise with high solitude ($\approx 76\%$ alone), combining discomfort with isolation.
Three methods converge	Fingerprint, lift, and residuals agree on the same behavioural ordering, strengthening confidence beyond any single test.
Effect sizes, not p-values, are the valid read-out	Responses are nested within 106 participants, so significance markers act as descriptive screens rather than formal tests.

4.1. Implications for sustainable urban design

The results of this analysis support the ability to close the loop between the design of how planners predict occupants behave and perceive their environment versus how they actually do in the real world. Much of the insights reinforce intuition, but there could be subregions that perform better or worse than expected. Heat and noise are two elements that are characterized, but numerous other subjective elements can be captured.

4.2. Limitations

Sample size and lack of diversity of participants make it difficult for these insights to be generalized across the population. Convenience sampling and lack or sensors.

4.3. Future work

Deeper characterization of the subset of the population that have certain parterns of behavior would help estabblish archetypes of behavior to identify which should be reinforced versus prevented from becoming ubiquitous. Targeted microsurveys based on where the person is and waht the subjective target of the deployment is. Development of larger deployments across people who have thier own smartwatch to result in possibly thousands or more participants.

5. Conclusion

[PLACEHOLDER: Concise restatement of the contribution and headline takeaway for sustainable urban development.]

Acknowledgements

The authors acknowledge the student researchers who assisted with data collection and processing: Charis Boey, Kristi Maisha, Annabel Sim, Aisyah Iskandar, Lum Jun Chao, and S Aravindkugesh.

CRediT authorship contribution statement

Clayton Miller: Visualization, Software, Resources, Methodology, Funding acquisition, Formal analysis, Writing – original draft, Supervision, Project administration, Investigation, Conceptualization. Mario Frei: Writing – review & editing, Validation, Project administration, Investigation, Conceptualization, Methodology, Data curation. Yun Xuan Chua: Methodology, Conceptualization, Writing – review & editing, Project administration, Investigation.

Declaration of generative AI use

The authors used Runcell AI, powered by OpenAI's GPT-5.6 Terra large language model, and Claude Opus 4.8 by Anthropic to assist with the organisation of the LaTeX manuscript, figure placement, and language polishing. These tools were not used to generate the study data, conduct the analysis, or replace the authors' scholarly judgement. All generated suggestions were reviewed and revised by the authors, who take full responsibility for the originality, accuracy, integrity, citations, and final content of this manuscript. Generative AI is not listed as an author.

Disclosure statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research was supported by the Singapore Ministry of Education (MOE) Tier 1 Grants titled *The Internet-of-Buildings (IoB) Platform – Testing of Visual Analytics for AI Technologies towards a Well and Green Built Environment* (A-0008305-01-00) and Ecological Momentary Assessment (EMA) for Built Environment Research (A0008301-01-00).

Data availability statement

The data and code supporting the findings of this study will be available on GitHub.

References

Abboushi, B., Safranek, S., Rodriguez-Feo Bermudez, E., Pratoomratana, S., Chen, Y., Poplawski, M., & Davis, R. (2022). A review of the use of wearables in indoor environmental quality studies and an evaluation of data accessibility from a wearable device. *Frontiers in Built Environment*, *8*.

Appel-Meulenbroek, R., Steps, S., Wenmaekers, R., & Arentze, T. (2020, January). Coping strategies and perceived productivity in open-plan offices with noise problems. *Journal of Managerial Psychology*, *36*(4), 400–414.

Barbour, E., Davila, C. C., Gupta, S., Reinhart, C., Kaur, J., & González, M. C. (2019, August). Planning for sustainable cities by estimating building occupancy with mobile phones. *Nature Communications*, *10*(1), 3736.

Castaldo, V. L., Pigliautile, I., Rosso, F., Cotana, F., De Giorgio, F., & Pisello, A. L. (2018, November). How subjective and non-physical parameters affect occupants' environmental comfort perception. *Energy and Buildings*, *178*, 107–129.

Christensen, T. C., Barrett, L. F., Bliss-Moreau, E., & Lebo, K. (2003, March). A practical guide to experience-sampling procedures. *Journal of Happiness Studies*, *4*(1), 53–78.

Fischer, T., Schraivogel, S., Caversaccio, M., & Wimmer, W. (2022, March). Are smartwatches a suitable tool to monitor noise exposure for public health awareness and otoprotection? *Frontiers in Neurology*, *13*, 856219.

Fuller, D., Shareck, M., & Stanley, K. (2017, October). Ethical implications of location and accelerometer measurement in health research studies with mobile sensing devices. *Social Science & Medicine*, *191*, 84–88.

Gharani, P., Karimi, H. A., Syzdykbayev, M., Burke, L. E., Rathbun, S. L., Davis, E. M., ... Mendez, D. D. (2021, June). Geographically-explicit ecological momentary assessment (GEMA) architecture and components: Lessons learned from PMOMS. *Informatics for Health and Social Care*, *46*(2), 158–177.

Gu, Y., Quintana, M., Liang, X., Ito, K., Yap, W., & Biljecki, F. (2025, August). Designing effective image-based surveys for urban visual perception. *Landscape and Urban Planning*, *260*, 105368.

Hammoud, R., Tognin, S., Smythe, M., Gibbons, J., Davidson, N., Bakolis, I., & Mechelli, A. (2024, April). Smartphone-based ecological momentary assessment reveals an incremental association between natural diversity and mental wellbeing. *Scientific Reports*, *14*(1), 7051.

Ignatius, M., Lim, J., Gottkehaskamp, B., Fujiwara, K., Miller, C., & Biljecki, F. (2024, June). Digital twin and wearables unveiling pedestrian comfort dynamics and walkability in cities. In *Isprs annals of the photogrammetry, remote sensing and spatial information sciences* (Vol. X-4/W5-2024, pp. 195–202).

Ito, K., Kang, Y., Zhang, Y., Zhang, F., & Biljecki, F. (2024, September). Understanding urban perception with visual data: A systematic review. *Cities*, *152*, 105169.

Kim, Y. (2016, January). Impacts of the perception of physical environments and the actual physical environments on self-rated health. *International Journal of Urban Sciences*, *20*(1), 73–87.

Kingsbury, C., Buzzi, M., Chaix, B., Kanning, M., Khezri, S., Kiani, B., ... Kestens, Y. (2024, June). STROBE-GEMA: A STROBE extension for reporting of geographically explicit ecological momentary assessment studies. *Archives of Public Health*, *82*(1), 84.

Kirchner, T. R., & Shiffman, S. (2016, September). Spatio-temporal determinants of mental health and well-being: Advances in geographically-explicit ecological momentary assessment (GEMA). *Social Psychiatry and Psychiatric Epidemiology*, *51*(9), 1211–1223.

Kumar, A., & Pandey, A. C. (2013, March). Spatio-temporal assessment of urban environmental conditions in ranchi township, india using remote sensing and geographical information system techniques. *International Journal of Urban Sciences*, *17*(1), 117–141.

Marquart, H., Schlink, U., & Nagendra, S. M. S. (2022, January). Complementing mobile measurements with walking interviews: A case study on personal exposure of commuters in chennai, india. *International Journal of Urban Sciences*, *26*(1), 148–161.

Miller, C., Abdelrahman, M., Chong, A., Biljecki, F., Quintana, M., Frei, M., … Wong, D. (2021, November). The internet-of-buildings (IoB): Digital twin convergence of wearable and IoT data with GIS/BIM. *Journal of Physics: Conference Series*, *2042*(1), 012041.

Miller, C., Chua, Y. X., Frei, M., & Quintana, M. (2022, December). Towards smartwatch-driven just-in-time adaptive interventions (JITAI) for building occupants. In *Proceedings of the 9th ACM international conference on systems for energy-efficient buildings, cities, and transportation* (pp. 336–339). New York, NY, USA: Association for Computing Machinery.

Miller, C., Chua, Y. X., Quintana, M., Lei, B., Biljecki, F., & Frei, M. (2025, October). Make yourself comfortable: Nudging urban heat and noise mitigation with smartwatch-based just-in-time adaptive interventions (JITAI). *Building and Environment*, *284*, 113388.

Miller, C., Ibrahim, M., Akbar, I. S., Picchetti, B., Chua, Y. X., Frei, M., … Fu, C. (2025, November). The cool, quiet city machine learning competition: Overview and results. *Journal of Physics: Conference Series*, *3140*(11), 112017.

Miller, C., Quintana, M., Frei, M., Chua, Y. X., Fu, C., Picchetti, B., … Biljecki, F. (2023, November). Introducing the cool, quiet city competition: Predicting smartwatch-reported heat and noise with digital twin metrics. In *Proceedings of the 10th ACM international conference on systems for energy-efficient buildings, cities, and transportation* (pp. 298–299). New York, NY, USA: Association for Computing Machinery.

Phuttharak, J., & Loke, S. (2020, January). Exploring incentive mechanisms for mobile crowdsourcing: Sense of safety in a thai city. *International Journal of Urban Sciences*, *24*(1), 13–34.

Quintana, M., Gu, Y., Liang, X., Hou, Y., Ito, K., Zhu, Y., … Biljecki, F. (2025, October). Global urban visual perception varies across demographics and personalities. *Nature Cities*, *2*(11), 1092–1106.

Reichert, M., Giurgiu, M., Koch, E., Wieland, L. M., Lautenbach, S., Neubauer, A. B., … Liao, Y. (2020, September). Ambulatory assessment for physical activity research: State of the science, best practices and future directions. *Psychology of Sport and Exercise*, *50*, 101742.

Schweiker, M., Rissetto, R., & Wagner, A. (2020, March). Thermal expectation: Influencing factors and its effect on thermal perception. *Energy and Buildings*, *210*, 109729.

Stone, A. A., & Shiffman, S. (1994, September). Ecological momentary assessment (EMA) in behavioral medicine. *Annals of Behavioral Medicine*, *16*(3), 199–202.

Tartarini, F., Frei, M., Schiavon, S., Chua, Y. X., & Miller, C. (2023, November). Cozie apple: An iOS mobile and smartwatch application for environmental quality satisfaction and physiological data collection. *Journal of Physics: Conference Series*, *2600*(14), 142003.

Wang, Z., de Dear, R., Luo, M., Lin, B., He, Y., Ghahramani, A., & Zhu, Y. (2018, June). Individual difference in thermal comfort: A literature review. *Building and Environment*, *138*, 181–193.

Yap, W., Stouffs, R., & Biljecki, F. (2023, July). Urbanity: Automated modelling and analysis of multidimensional networks in cities. *npj Urban Sustainability*, *3*(1), 1–11.

Yıldırım, O. C., Heyik, M. A., Güleç Ozer, D., & Sungur, A. (2026, July). Measuring perceived walkability in istanbul's historic peninsula: A multi-method integration of PPGIS, EEG, and self-reported emotions. *International Journal of Urban Sciences*, 1–31.

Zhang, X., Zhou, S., Kwan, M.-P., Su, L., & Lu, J. (2020, November). Geographic ecological momentary assessment (GEMA) of environmental noise annoyance: The influence of activity context and the daily acoustic environment. *International Journal of Health Geographics*, *19*(1), 50.